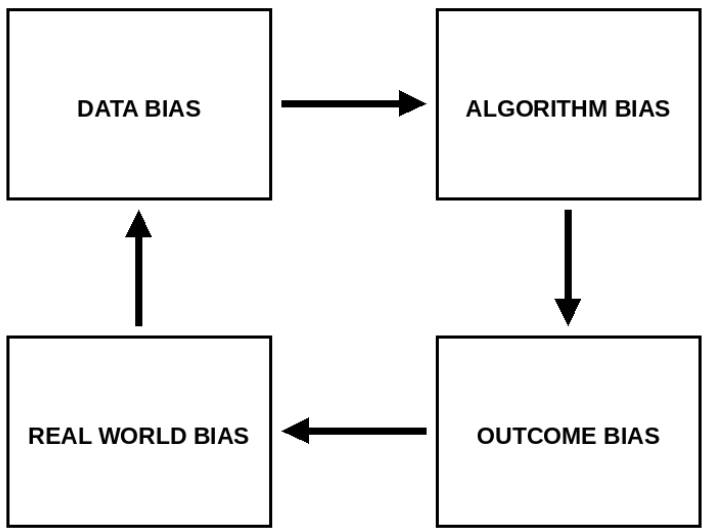
PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

November 11, 2026

Bias



Central points:

- It starts with *people*...
- $\rightarrow$ data...
- Modeling / outcome choices
- Accuracy $\neq$ fairness
- Impacts aren't symmetrical
- Intent?...

Case Study #1: Healthcare

From Johnson (2019):

- **The Algorithm**: Optum's tool for flagging patients for intensive care management
- **The Study**: Obermeyer et al. (*Science*, 2019)
- **The Bias**: Algorithm underestimated the health needs of the sickest Black patients
- **The Mechanisms**:
  - Proxy: Outcome = healthcare *costs*, not health
  - Traits: The algorithm *didn't include race as a predictor* (and it still resulted in bias)
  - Asymmetry: bias reduced the proportion of Black patients identified for the care management program by more than half

Case Study #2: Criminal Recidivism

From Angwin et al. (2016):

- **The Algorithm**: Northpointe's COMPAS recidivism tool
- **The Study**: *Pro Publica* (2016): 7000 COMPAS scores
- **The Bias**: Algorithm over-predicted risk for Black defendants and under-predicted risk for white defendants
- **The Mechanisms**:
  - Traits (again): Race *was not predictor*
  - Subgroups:
    - false positives: 44.9% among Blacks, 23.5% for whites
    - false negatives: 47.7% among whites, 28.0% for Blacks
- **The Defense**: Calibrated accuracy was identical across the two groups

Case Study #3: Employment

From Milne (2024):

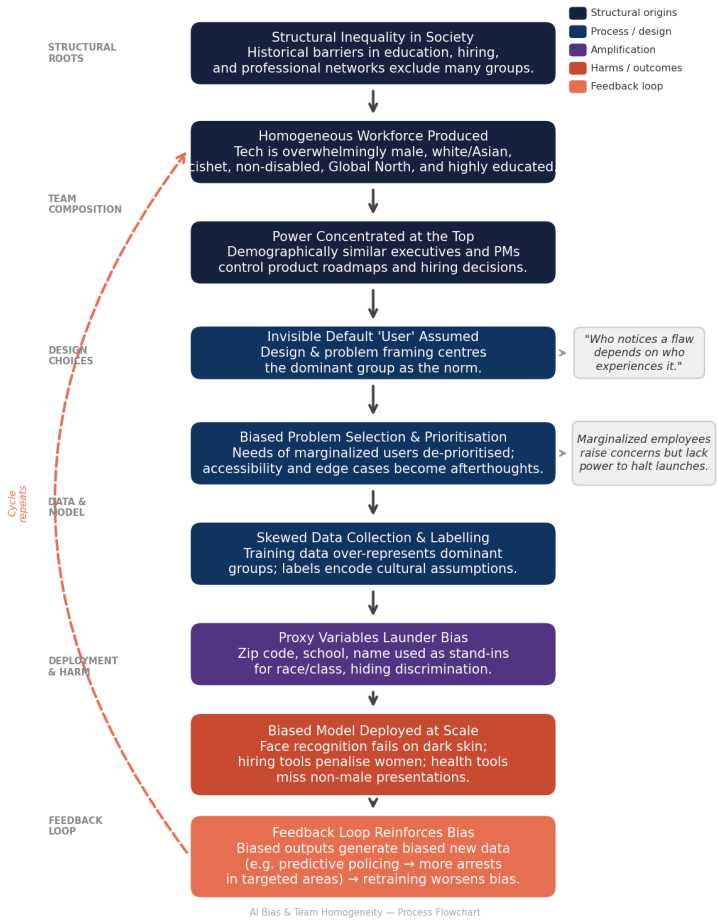
- **The Algorithm**: State-of-the-art LLMs from three large employment companies
- **The Study**: Wilson and Calisklan (*AIES* 2024)
- **The Bias**: Audit study: Across 500 listings in nine occupations, LLMs:
  - Preferred white names 85.1% of the time
  - Intersectional: Black male names faced the steepest penalty; Black female names were sometimes preferred over Black male names but rarely over white
- **The Mechanisms**:
  - Traits: only implicit (via names)
  - The Data

Case Study #4: Employment Redux

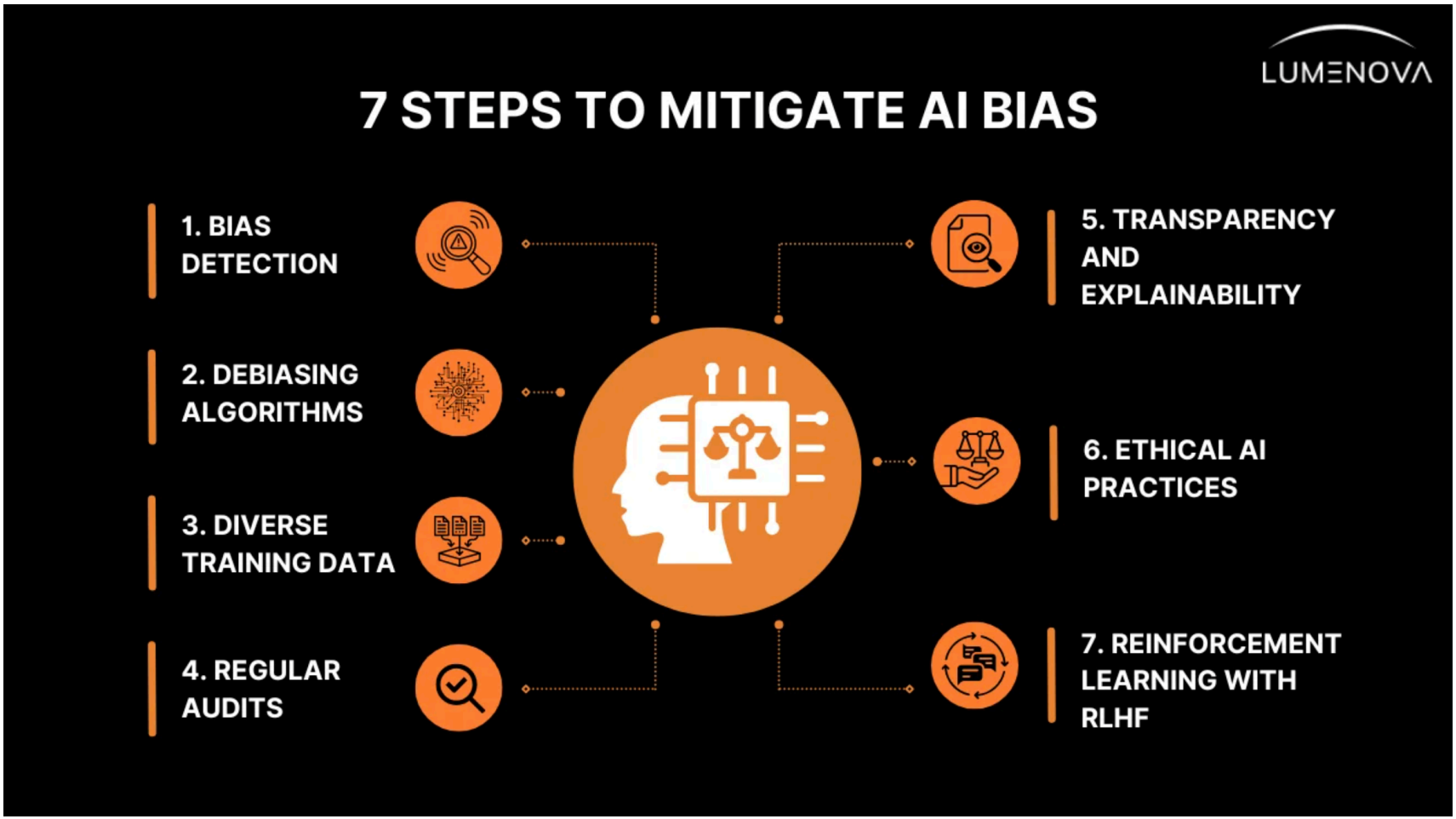
From Gupta (2025):

- **The Algorithm**: 1.4m web images + nine LLMs
- **The Study**: Guilbeault et al. (*Nature* 2025)
- **The Bias**: ChatGPT made female candidates ≈1.6 years younger than male candidates, then ranked those younger-seeming female resumes less qualified
- **The Mechanisms**:
  - The Data: Upstream images tend to show women as younger
  - Feedback: Bias is self-reinforcing
  - Groups: Fixing one bias may cause another

Mechanisms



- "Defaults"
- Organizational power: agenda-setting, funding, decision-making
- Noticing problems
- Deciding what gets fixed
- Data $\rightarrow$ Algorithmic bias
- Feedback loops


LUMENOVA
7 STEPS TO MITIGATE AI BIAS
1. BIAS DETECTION
2. DEBIASING ALGORITHMS
3. DIVERSE TRAINING DATA
4. REGULAR AUDITS
5. TRANSPARENCY AND EXPLAINABILITY
6. ETHICAL AI PRACTICES
7. REINFORCEMENT LEARNING WITH RLHF